Name______________________________

The Remainder Theorem and Bounds

Date________________ Period____

Evaluate $f(x)$ at k.

1) $f(x) = x^3 - 4x^2 + 7x$
$k = -1$

2) $f(x) = x^3 - 16x$
$k = 1$

3) $f(x) = x^4 + 4x^3 + 5x^2 - 4x - 4$
$k = 1$

4) $f(x) = 5x^5 - 4x^4 - 5x^2 - 5$
$k = 1$

Find the remainder when $f(x)$ is divided by $x - k$.

5) $f(x) = 5x^6 + 5x^5 - 7x^4 + 7x^3 + 5x^2 - 4x - 1$
$k = -2$

6) $f(x) = 5x^4 - 6x^3 - x^2 + 7x - 2$
$k = 2$

7) $f(x) = x^4 + 12x^3 + 37x^2 + 42x + 16$
$k = -3$

8) $f(x) = 4x^6 - 5x^5 - 9x^4 + 2x^2 + 3x - 9$
$k = -1$